FOR A MORPHISM OF NOETHERIAN SCHEMES, F_* ON QUASI-COHERENT SHEAVES ADMITS A RIGHT DERIVED EXTENSION COMPATIBLE WITH RF_*

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Editorial note. Inputs lemma-acyclicity-lemma and lemma-injective-quasi-coherent-sheaf-Noetherian are same-chapter NOT carded (cross-line-claimed labels) and are declared external.

Allowed standard prerequisites (editorial).

- Noetherian morphisms are qcqs. Properties, lemma locally-Noetherian-quasi-separated; Schemes, remark quasi-compact-and-quasi-separated (cross-chapter, named). f_* preserves quasi-coherence.
- K-injective complexes of injectives in QCoh. Injectives, theorem K-injective-embedding-grothendieck (cross-chapter, named). Represents K by a K-injective complex with injective terms, defining Phi.
- Acyclicity criterion for the canonical map. lemma-acyclicity-lemma (same chapter, NOT carded, declared external). f_*I-bullet -> Rf_*I-bullet is an isomorphism once the terms are f_*-acyclic.
- Injective quasi-coherent sheaves on Noetherian schemes are flasque. lemma-injective-quasi-coherent-sheaf-Noetherian (same chapter, NOT carded, declared external). Supplies the flasqueness of the terms.
- Flasque sheaves are acyclic for pushforward. Cohomology, lemma flasque-acyclic-pushforward (cross-chapter, named). Concludes each term is right f_*-acyclic.

Source reference. Original reference 09T0: focused-seeds/geometry/S9-perf2-Noetherian-pushforward/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 079P: focused-seeds/geometry/S9-perf2-Noetherian-pushforward/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G9N: sources/stacks-perfect/pinned-original-source.tex, source lines 731–731. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 09T2: sources/stacks-perfect/pinned-original-source.tex, source lines 1952–1952. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01OY: focused-seeds/geometry/S9-perf2-Noetherian-pushforward/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 077P: focused-seeds/geometry/S9-perf2-Noetherian-pushforward/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01LC: focused-seeds/geometry/S9-perf2-Noetherian-pushforward/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0816: focused-seeds/geometry/S9-perf2-Noetherian-pushforward/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of Noetherian schemes. Then f_* on quasi-coherent sheaves has a right derived extension $\Phi : D(QCoh(\mathcal{O}_X)) \to D(QCoh(\mathcal{O}_Y))$ such that the diagram*

$$\begin{array}{ccc} D(QCoh(\mathcal{O}_X)) & \longrightarrow & D_{QCoh}(\mathcal{O}_X) \\ \downarrow{\scriptstyle \Phi} & & \downarrow{\scriptstyle Rf_*} \\ D(QCoh(\mathcal{O}_Y)) & \longrightarrow & D_{QCoh}(\mathcal{O}_Y) \end{array}$$

commutes.

Proof. Since X and Y are Noetherian schemes the morphism is quasi-compact and quasi-separated (see Properties, Lemma 01OY and Schemes, Remark 0816). Thus f_* preserve quasi-coherence, see Schemes, Lemma 01LC. Next, let K be an object of $D(QCoh(\mathcal{O}_X))$. Since $QCoh(\mathcal{O}_X)$ is a Grothendieck abelian category (Properties, Proposition 077P), we can represent K by a K-injective complex $\mathcal{I}^\bullet$ such that each $\mathcal{I}^n$ is an injective object of $QCoh(\mathcal{O}_X)$, see Injectives, Theorem 079P. Thus we see that the functor Φ is defined by setting

$$\Phi(K) = f_*\mathcal{I}^\bullet$$

where the right hand side is viewed as an object of $D(QCoh(\mathcal{O}_Y))$. To finish the proof of the lemma it suffices to show that the canonical map

$$f_*\mathcal{I}^\bullet \longrightarrow Rf_*\mathcal{I}^\bullet$$

is an isomorphism in $D(\mathcal{O}_Y)$. To see this by Lemma 0G9N it suffices to show that $\mathcal{I}^n$ is right f_*-acyclic for all $n \in \mathbf{Z}$. This is true because $\mathcal{I}^n$ is flasque by Lemma 09T2 and flasque modules are right f_*-acyclic by Cohomology, Lemma 09T0. $\square$